

More Than Words: Quantifying Language to Measure Firms' Fundamentals

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ABSTRACT

We examine whether a simple quantitative measure of language can be used to predict individual firms' accounting earnings and stock returns. Our three main findings are: (1) the fraction of negative words in firm-specific news stories forecasts low firm earnings; (2) firms' stock prices briefly underreact to the information embedded in negative words; and (3) the earnings and return predictability from negative words is largest for the stories that focus on fundamentals. Together these findings suggest that linguistic media content captures otherwise hard-to-quantify aspects of firms' fundamentals, which investors quickly incorporate into stock prices.

Language is conceived in sin and science is its redemption

—W. V. Quine, *The Roots of Reference*

A VOLUMINOUS LITERATURE EXAMINES the extent to which stock market prices incorporate quantitative information. Although few researchers study the impact of qualitative verbal information, there are compelling theoretical and empirical reasons to do so.¹ Theoretically, efficient firm valuations should be equal to the expected present discounted value of their cash flows conditional on investors' information sets, which include qualitative descriptions of firms' business environments, operations, and prospects in the financial press. Empirically, substantial movements in firms' stock prices do not seem to correspond to changes in quantitative measures of firms' fundamentals (e.g., Shiller (1981), Roll (1988), and Cutler, Poterba, and Summers (1989)), suggesting that qualitative variables may help explain stock returns.

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¹ In Section I, we discuss several recent studies that examine qualitative verbal information.

In this paper we quantify the language used in financial news stories in an effort to predict firms' accounting earnings and stock returns. Our study takes as a starting point Tetlock (2007), who examines how qualitative information—in particular, the fraction of negative words in a widely read news column about the stock market—is incorporated in aggregate market valuations. We extend that analysis to address the impact of negative words in all *Wall Street Journal* (WSJ) and *Dow Jones News Service* (DJNS) stories about individual S&P 500 firms from 1980 to 2004.² In addition to studying individual firms' stock returns, we investigate whether negative words can be used to improve expectations of firms' future cash flows. Overall, this study sheds light on whether and why quantifying language provides novel information about firms' earnings and returns.

Before delving into our tests, we call attention to two significant advantages to using the language in everyday news stories to predict firms' earnings and returns. First, by quantifying language, researchers can examine and judge the directional impact of a limitless variety of events, whereas most studies focus on one particular event type, such as earnings announcements, mergers, or analysts' recommendations. Analyzing a more complete set of events that affect firms' fundamental values allows researchers to identify common patterns in firm responses and market reactions to events. Equally important, examining all newsworthy events simultaneously limits the scope for “dredging for anomalies”—the phrase used by Fama (1998) to describe running event studies on different types of events until one obtains “significant” results.

Second, linguistic communication is a potentially important source of information about firms' fundamental values. Because very few stock market investors directly observe firms' production activities, they get most of their information secondhand. Their three main sources are analysts' forecasts, quantifiable publicly disclosed accounting variables, and linguistic descriptions of firms' current and future profit-generating activities. If analyst and accounting variables are incomplete or biased measures of firms' fundamentals, linguistic variables may have incremental explanatory power for firms' future earnings and returns.

As an example of our linguistic quantification method, consider a January 8, 1999 DJNS article entitled “Consumer Groups Say Microsoft Has Overcharged for Software.” We hypothesize that the fraction of negative words contained in the article is related to the impact of the news event on Microsoft's market value (Tetlock (2007)). The article's second sentence is: “The alleged ‘pricing abuse will only get worse if Microsoft is not disciplined sternly by the antitrust court,’ said Mark Cooper, director of research for Consumer Federal of America.” Based on the classification dictionary that we use, this sentence's fraction of negative words ranks in the 99th percentile of sentences within our news

² As in Tetlock (2007), we use negative words from the General Inquirer's Harvard-IV-4 classification dictionary to measure qualitative information. Our results are similar for alternative measures that include positive words from this same dictionary. See Section II for further discussion.

database.³ In this case, the abundance of negative words is consistent with an intuitive reading of the story, and with Microsoft's abnormally poor stock returns around the news event.⁴

We do not claim that our crude quantitative measure of language subsumes or dominates traditional accounting measures of firms' fundamentals. Rather, we investigate whether the fraction of negative words in firm-specific news stories can improve our understanding of firms' cash flows and whether firms' stock market prices efficiently incorporate linguistic information. Insofar as negative word counts are noisy measures of qualitative information, the coefficients in our regressions should be biased toward zero, understating the true importance of qualitative information.

Despite this large measurement error, our first main result is that negative words convey negative information about firm earnings above and beyond stock analysts' forecasts and historical accounting data. In other words, qualitative verbal information does not merely echo easily quantifiable traditional measures of firm performance. We also test whether stock market prices rationally reflect the effect of negative words on firms' expected earnings. Our second result is that stock market prices respond to the information embedded in negative words with a small, one-day delay. As a result, we identify potential profits from using daily trading strategies based on the words in a continuous intraday news source (*DJNS*), but not from strategies based on a news source updated less frequently (*WSJ*). Accounting for reasonable transaction costs could eliminate the profitability of the high-frequency trading strategy, suggesting that short-run frictions play an important role in how information is incorporated in asset prices. To interpret these results further, we separately analyze negative words in news stories whose content focuses on firms' fundamentals. We find that negative words in stories about fundamentals predict earnings and returns more effectively than negative words in other stories. Collectively, our three findings suggest that linguistic media content captures otherwise hard-to-quantify aspects of firms' fundamentals, which investors quickly incorporate into stock prices.

The layout of the paper is as follows. In Section I we conduct a brief review of related research on qualitative information. Section II discusses the properties of the news stories used in this study. Sections III and IV present the main tests for whether negative words predict firms' earnings and stock returns, respectively. In Section V, we assess whether earnings and return predictability is strongest for timely (*DJNS*) news articles that focus on firms' fundamentals. In Section VI, we present our conclusions and outline directions for further research on media content.

³ There are five negative words (alleged, abuse, worse, sternly, and antitrust) among the 29 total words in the sentence, or 17.2%, which exceeds the cutoff for the 99th percentile of our 1998 news story data. The tone of the sentence is representative of the entire article, which also ranks in the top decile for 1998.

⁴ Microsoft's cumulative abnormal stock returns were -42, -141, and -194 basis points for the three trading days surrounding the news event.

I. Research on Qualitative Information

To create a quantitative variable from text documents such as news stories, one must devise a representation of the unstructured text. The most common representation is the Bag-of-Words scheme, which represents all words appearing in news stories as a document-term matrix—for example, a row could be the 1/8/99 Microsoft story above, and columns could be the terms “alleged,” “abuse,” “worse,” “happy,” and “neutral.” The matrix elements are designed to capture the information value of each word in each news story, which could be the relative frequencies of the 5 words within the 29-word excerpt: [1/29, 1/29, 1/29, 0/29, 0/29]. The challenge in text analysis is to translate this term–document matrix into a meaningful conceptual representation of the story, such as the degree to which the story conveys positive or negative information.

In this paper, we collapse the document-term matrix into just two columns using domain knowledge from the positive and negative word categories in the Harvard-IV-4 psychosocial dictionary. For reasons explained below, our primary focus is the negative column. We make the simplifying assumption that all negative words in the predetermined dictionary are equally informative, and other words are uninformative. As in the example above, we measure a story’s negativity according to the relative frequency of negative words in each news story. These procedures conform to Tetlock (2007) and many psychological studies using the Harvard-IV-4 dictionary. A well-known and widely used text analysis program called the General Inquirer features this same dictionary.⁵

A more sophisticated alternative to our approach would entail estimating the information value of each word’s occurrence in a story, and determining which words are most likely to appear in negative stories. Unfortunately, these nuances have significant drawbacks. First, subjective human judgment may be necessary to assess whether a story is negative. Second, determining which words are more likely to have negative meanings requires the estimation of potentially thousands of likelihoods ratios—one for every word used in classification. By contrast, we rely on extensive psychological research to identify negative words, thereby avoiding this daunting estimation task and the need for subjective human judgment. Our resulting word count measures are parsimonious, objective, replicable, and transparent. At this early stage in research on qualitative information, these four attributes are particularly important, and give word count measures a reasonable chance of becoming widely adopted in finance.

In addition to Tetlock (2007), several new research projects investigate the importance of qualitative information in finance. Our study is most closely related to concurrent work by Li (2006) and Davis, Piger, and Sedor (2006), who analyze the tone of qualitative information using objective word counts from corporate annual reports and earnings press releases, respectively. Whereas

⁵ The Harvard-IV-4 dictionary on the General Inquirer’s Web site lists each word in the negative category: <http://www.webuse.umd.edu:9090/tags/TAGNeg.html>. See Riffe, Lacy, and Fico (1998) for a survey of content analysis and its application to the media.

Davis, Piger, and Sedor (2006) examine the contemporaneous relationships between earnings, returns, and qualitative information, Li (2006) focuses on the predictive ability of qualitative information as we do.

Li (2006) finds that the two words “risk” and “uncertain” in firms’ annual reports predict low annual earnings and stock returns, which the author interprets as underreaction to “risk sentiment.” Our study differs from Li (2006) in that we examine qualitative information in news stories at daily horizons rather than qualitative information in annual reports at annual horizons. Our predictability tests use over 80 quarters of earnings and over 6,000 days of returns data, as compared to 12 years of earnings and 12 years of returns data in Li (2006). Other differences between our studies, such as the measures used, do not seem to be as important. When we use the words “risk” and “uncertain” rather than the entire negative words category to measure qualitative information, we find similar albeit slightly weaker earnings and return predictability.

Some prior and contemporaneous research analyzes qualitative information using more sophisticated subjective measures, rather than simple objective word counts. However, most of this work focuses on firms’ stock returns and ignores firms’ earnings. For example, Antweiler and Frank (2004) and Das and Chen (2006) design algorithms to reproduce humans’ “bullish,” “neutral,” or “bearish” ratings of internet chat room messages and news stories. Neither study finds any statistically significant return predictability in individual stocks. A recent study by Antweiler and Frank (2006), which uses an algorithm to identify news stories by their topic rather than their tone, does find some return predictability. For many of their topic classifications, Antweiler and Frank (2006) find significant return reversals in the 10-day period around the news, which they interpret as overreaction to news regardless of its tone.

II. Stylized Facts about Firm-Specific News Stories

We concentrate our analysis on the fraction of negative words in *DJNS* and *WSJ* stories about S&P 500 firms from 1980 through 2004. We choose the S&P 500 constituent firms for reasons of importance and tractability. Firms in the S&P 500 index encompass roughly three-quarters of the total U.S. market capitalization, and appear in the news sufficiently often to make the analysis interesting.

We obtain S&P index constituents and their stock price data from the Center for Research on Security Prices (CRSP), analyst forecast information from the Institutional Brokers’ Estimate System (I/B/E/S), and accounting information from Compustat. Merging the news stories and the financial information for a given firm requires matching firms’ common names used in news stories to their permnos, CUSIPs, or gvkeys used in the above financial data sets. Although firms’ common names usually resemble the firm names appearing in financial data sets, perfect matches are rare.

To obtain the common names that we use as search strings for news stories, we begin with the company name variable in the CRSP data for all S&P 500 index constituents during the relevant time frame. We use the CRSP company

name change file to identify situations in which a firm in the index changes its name. We focus on news stories featuring the company name most directly related to the stock. Thus, for conglomerates, we use the holding company name, not the subsidiary names—for example, PepsiCo, Inc., or Pepsi for short, rather than Gatorade or Frito-Lay. This means that we may miss news stories about some firms' major products, possibly weakening our results.

Our source for news stories is the Factiva database. To find the name that media outlets use to refer to a firm, we use a combination of four different methods that are described in detail in the Appendix. Because of the large number of firms and news stories, we implement an automated story retrieval system. For each S&P 500 firm, the system constructs a query that specifies the characteristics of the stories to be retrieved. The system then submits the query and records the retrieved stories.

In total, we retrieve over 350,000 qualifying news stories—over 260,000 from *DJNS* and over 90,000 from *WSJ*—that contain over 100,000,000 words. We find at least one story for 1,063 of 1,110 (95.8%) of the firms in the S&P 500 from 1980 to 2004 (see the Appendix for details). We include a news story in our analysis only if it occurs while the firm is a member of the S&P index and is within our 25-year time frame. We also exclude stories in the first week after a firm has been newly added to the index to prevent the well-known price increase associated with a firm's inclusion in the S&P 500 index from affecting our analysis (Shleifer (1986)).

Each of the stories in our sample meets certain requirements that we impose to eliminate irrelevant stories and blurbs. Specifically, we require that each firm-specific story mentions the firm's official name at least once within the first 25 words, including the headline, and the firm's popular name at least twice within the full story. In addition, we require that each story contains at least 50 words in total, and at least 5 words that are either "Positive" or "Negative," where at least 3 of the 5 must be unique. We impose these three word count filters to eliminate stories that contain only tables or lists with company names and quantitative information, and to limit the influence of outliers on the negative words measure described below.

Following Tetlock (2007), our primary measure of media content is the standardized fraction of negative words in each news story. In unreported tests, we find very similar results using combined measures of positive (P) and negative (N) words, such as $(P - N)/(P + N)$ and $\log((1 + P)/(1 + N))$. However, using positive words in isolation produces much weaker results, especially after controlling for negative words. These results are consistent with the general analysis of word categories in Tetlock (2007). That study shows that negative words summarize common variation in the entire set of General Inquirer word categories better than any other single category, including positive words—that is, negatives are most highly correlated with the first eigenvector of the N by N variance-covariance matrix for all N word categories. Tetlock (2007) also finds that negative words have a much stronger correlation with stock returns than other words. These results are also consistent with a large body of literature in psychology—for example, Baumeister et al. (2001) and Rozin and Royzman (2001)—that argues negative information has more impact and

is more thoroughly processed than positive information across a wide range of contexts.

Before counting instances of negative words, we combine all qualifying news stories for each firm on a given trading day into a single composite story. We standardize the fraction of negative words in each composite news story by subtracting the prior year's mean and dividing by the prior year's standard deviation of the fraction of negative words. Formally, we define two measures of negative words:

$$Neg = \frac{\text{No. of negative words}}{\text{No. of total words}} \quad (1)$$

$$neg = \frac{Neg - \mu_{Neg}}{\sigma_{Neg}}, \quad (2)$$

where μ_{Neg} is the mean of Neg and σ_{Neg} is the standard deviation of Neg over the prior calendar year. The standardization may be necessary if Neg is non-stationary, which could happen if there are regime changes in the distribution of words in news stories—for example, the *DJNS* or *WSJ* changes its coverage or style. The variable neg is the stationary measure of media content that we employ in our regression analyses.

Before analyzing the predictive power of linguistic media content, we document an important stylized fact: There are many more firm-specific news stories in the days immediately surrounding a firm's earnings announcement. For each firm-specific news story, we calculate the number of days until the firm's next earnings announcement and the number of days that have passed since the firm's previous earnings announcement. We plot a histogram of both variables back-to-back in Figure 1. Thus, each story is counted exactly twice in Figure 1, once after the previous announcement and once before the next announcement, except the stories that occur on the earnings announcement day.

Figure 1 provides striking evidence that news stories concentrate around earnings announcement days, as shown by the three adjacent spikes representing the firm-specific news stories one day before, on the same day as, and one day after a firm's earnings announcement. This finding suggests that news stories could play an important role in communicating and disseminating information about firms' fundamentals. In the next three sections, we provide further support for this interpretation of Figure 1.

III. Using Negative Words to Predict Earnings

We now formally investigate whether the language used by the media provides new information about firms' fundamentals and whether stock market prices efficiently incorporate this information. In order to affect stock returns, negative words must convey novel information about either firms' cash flows or investors' discount rates (Campbell and Shiller (1987)). Our tests in this section focus on whether negative words can predict earnings, a proxy for cash flows,

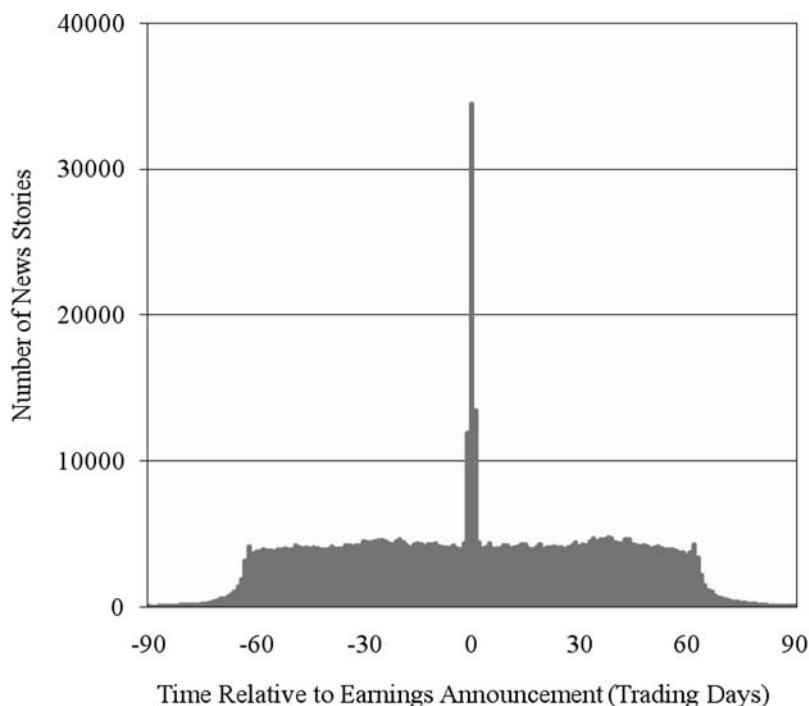


Figure 1. Media coverage around earnings announcements. This figure depicts the relationship between the number of firm-specific news stories and the number of days away from a firm's earnings announcement. All stories included in the figure are about S&P 500 firms, appear in either *Dow Jones News Service* or *The Wall Street Journal* from 1980 through 2004, and meet basic minimum word requirements (see text for details). For each news story, we calculate the number of days until the firm's next earnings announcement and the number of days that have passed since the firm's last earnings announcement. We plot a histogram of both variables back-to-back in Figure 1. Thus, each story is counted twice in Figure 1, once before and once after the nearest announcement, except the stories occurring on the earnings announcement day.

and therefore permanent changes in prices. The return predictability tests in Section IV address the possibility that negative words proxy for changes in investors' discount rates, and therefore lead to return reversals. The idea underlying our earnings predictability tests is that negative words in a firm's news stories prior to the firm's earnings announcement could measure otherwise hard-to-quantify unfavorable aspects of the firm's business environment.

We use two measures of firms' quarterly accounting earnings as dependent variables in our predictability tests, as the quarterly frequency is the highest frequency for earnings data. Our main tests compute each firm's standardized unexpected earnings (*SUE*) following Bernard and Thomas (1989), who use a seasonal random walk with trend model for each firm's earnings:

$$UE_t = E_t - E_{t-4} \quad (3)$$

$$SUE_t = \frac{UE_t - \mu_{UE_t}}{\sigma_{UE_t}}, \quad (4)$$

where E_t is the firm's earnings in quarter t , and the trend and volatility of unexpected earnings (UE) are equal to the mean (μ) and standard deviation (σ) of the firm's previous 20 quarters of unexpected earnings data, respectively. As in Bernard and Thomas (1989), we require that each firm have nonmissing earnings data for the most recent 10 quarters and assume a zero trend for all firms with fewer than 4 years of earnings data.

We also use standardized analysts' forecast errors ($SAFE$) as an alternative measure of firms' earnings to ensure robustness. $SAFE$ is equal to the median stock analyst's earnings forecast error divided by earnings volatility (σ), which is the same as the denominator of SUE . We use the median analyst forecast from the most recent statistical period in the I/B/E/S summary file prior to 3 days before the earnings announcement.⁶ We winsorize SUE and all analyst forecast variables at the 1% level to reduce the impact of estimation error and extreme outliers, respectively. Despite the well-known biases in stock analysts' earnings forecasts, we find remarkably similar results using SUE and $SAFE$.⁷

We attempt to match the frequency of our news measure to the frequency of our quarterly earnings variable. Our measure of negative words ($neg_{-30,-3}$) is the standardized number of negative words in all news stories between 30 and 3 trading days prior to an earnings announcement divided by the total number of words in these news stories. That is, we construct the measure exactly analogous to the story-specific measure (neg) defined earlier, where we treat all the words in the $[-30,-3]$ time window as though they form a single composite news story. We standardize $neg_{-30,-3}$ by subtracting the prior year's mean and dividing by the prior year's standard deviation.

The timing of $neg_{-30,-3}$ is designed to include news stories about the upcoming quarter's earnings announcement. Because 30 trading days is roughly one-half of a trading quarter, it is likely that most of the news stories in the $[-30,-3]$ time window focus on the firm's upcoming announcement rather than its previous quarter's announcement. In addition, we allow for 2 full trading days between the last news story included in this measure and the earnings announcement because Compustat earnings announcement dates may not be exact. None of our qualitative results change if we set the beginning of the time window to 20 or 40 trading days before the announcement, or set the ending of the window to 1 or 5 trading days before the announcement.

In all earnings predictability regressions, we include control variables based on a firm's lagged earnings, size, book-to-market ratio, trading volume, three measures of recent stock returns, analysts' earnings forecast revisions, and analysts' forecast dispersion. We measure firms' lagged earnings using last quarter's SUE or $SAFE$ measure, depending on which of these two variables is

⁶ Based on our conversations with Wharton Research Data Services representatives, the median forecast comes from the distribution that includes only the most up-to-date forecasts from each brokerage.

⁷ Several studies argue that analyst earnings forecasts are too optimistic (e.g., Easterwood and Nutt (1999)), overreact to certain pieces of information (e.g., De Bondt and Thaler (1990)), and underreact to other information (e.g., Abarbanell and Bernard (1992)), among other biases.

the dependent variable in the regression.⁸ We measure firm size ($\text{Log}(\text{Market Equity})$) and book-to-market ($\text{Log}(\text{Book}/\text{Market})$) at the end of the preceding calendar year, following Fama and French (1992). We compute trading volume as the log of annual shares traded divided by shares outstanding ($\text{Log}(\text{Share Turnover})$) at the end of the preceding calendar year.

Our three control variables for a firm's past returns are based on a simple earnings announcement event study methodology.⁹ We estimate benchmark returns using the Fama-French (1993) three-factor model with an estimation window of $[-252, -31]$ trading days prior to the earnings announcement. We include two control variables for a firm's recent returns, the cumulative abnormal return from the $[-30, -3]$ trading day window ($\text{FFCAR}_{-30, -3}$) and the abnormal return on day -2 ($\text{FFCAR}_{-2, -2}$). These return windows end 1 trading day after our $[-30, -3]$ news story time window to ensure that we capture the full price impact of the news stories. Our third control variable, $\text{FFAlpha}_{-252, -31}$, is the estimated intercept from the event study regression that spans the $[-252, -31]$ time window. We interpret the $\text{FFAlpha}_{-252, -31}$ measure as the firm's in-sample cumulative abnormal return over the previous calendar year, skipping the most recent month. The $\text{FFAlpha}_{-252, -31}$ variable is related to the Jegadeesh and Titman (1993) return momentum effect, which is based on firms' relative returns over the previous calendar year excluding the most recent month.

In all our earnings regressions, we include control variables for the median analyst's quarterly forecast revision and analysts' quarterly forecast dispersion. We compute the median analyst's 3-month earnings forecast revision (*Forecast Revisions*) following Chan, Jegadeesh, and Lakonishok (1996). We use 3-month revision periods rather than 6-month periods because these revisions capture new information after the forecast preceding last quarter's earnings announcement, which is already included in our regressions as a separate control. This revision variable is equal to the 3-month sum of scaled changes in the median analyst's forecast, where the scaling factor is the firm's stock price in the prior month. We compute analysts' forecast dispersion (*Forecast Dispersion*) as the standard deviation of analysts' earnings forecasts in the most recent time period prior to the announcement scaled by earnings volatility (σ)—that is, the denominator of *SUE* and *SAFE*. We construct both of these control variables using quarterly analyst forecasts to match our dependent variables, which are based on quarterly earnings measures. Because analysts' quarterly forecasts are unavailable from I/B/E/S between 1980 and 1983 and for firms without analyst coverage, the earnings predictability regressions that we report do not include these observations.¹⁰

Even though the stock return control variable ($\text{FFCAR}_{-30, -3}$) includes all of the information embedded in news stories during the $[-30, -3]$ time window, it

⁸ The inclusion of additional lags of the dependent variables does not change the results.

⁹ Controlling for alternative measures of past returns such as raw event returns and the past calendar year's return does not change our qualitative results.

¹⁰ If we omit the two analyst variables and include these remaining observations in our regressions, we find very similar results.

is possible that these stories are more recent than the most recent analyst forecast data. Indeed, many *WSJ* and *DJNS* news stories explicitly mention stock analysts, suggesting negative words in these stories may draw some predictive power from analysts' qualitative insights. To guard against the possibility that negative words predict returns solely because they appear more recently than the *quantitative* analyst forecasts, we also calculate a "Before Forecasts" negative words measure ($neg_{-30,-3}$) that includes only the stories that occur at least 1 trading day prior to the date of the most recent consensus analyst forecast.¹¹

We estimate the ability of negative words ($neg_{-30,-3}$) to predict earnings (*SUE* or *SAFE*) using pooled ordinary least squares (OLS) regressions and standard errors clustered by calendar quarter (Froot (1989)). The rationale is that the dependent variable (*SUE* or *SAFE*) is already standardized to remove any firm effect, but does exhibit a time effect because firms' realized earnings are undoubtedly correlated within calendar quarters. We confirm this reasoning using several diagnostic checks following Petersen (2007). These tests also suggest that, if anything, our coefficients and standard errors are conservative relative to a wide range of alternative estimation techniques.¹²

Table I reports estimates of the ability of negative words ($neg_{-30,-3}$) to predict quarterly earnings using six OLS regressions: two different dependent variables (*SUE*; *SAFE*) regressed on negative words computed based on four sets of news stories (*DJNS*, *WSJ*, "Before Forecasts," and "All Stories"). The key result is that negative words ($neg_{-30,-3}$) consistently predict lower earnings, regardless of whether we use the *SUE* or *SAFE* measure, and regardless of whether we use stories from *DJNS* or *WSJ* or from the time period before stock analysts state their earnings forecasts.

Although negative words ($neg_{-30,-3}$) from *WSJ* stories appear to predict *SUE* slightly better than $neg_{-30,-3}$ from *DJNS* stories, the *WSJ* coefficient estimates of $neg_{-30,-3}$ are not statistically different from the *DJNS* estimates. All six estimates of the dependence of earnings on negative words are negative and statistically significant at the 99% level. Because the independent and dependent variables are standardized, the rough economic interpretation of the "All Stories" *SUE* estimate is that the conditional expectation of *SUE* is $4 \times (0.064) = 0.255$ standard deviations lower as $neg_{-30,-3}$ increases from two standard deviations below to two standard deviations above its mean value.

We now analyze the *SUE* and *SAFE* regressions that compute negative words using stories from both news sources in greater detail. Columns 4 and 6 in

¹¹ Because I/B/E/S reviews and updates the accuracy and timing of analyst forecasts even after the consensus forecast date, it is unlikely that news stories from one trading day earlier contain information not reflected in the consensus. In addition, allowing three trading days does not change our qualitative results.

¹² If we use fixed-effects models instead, the point estimates of the key coefficients increase and the standard errors remain similar. This robustness is comforting because fixed-effects estimators and pooled OLS estimators for dynamic panel data models with lagged dependent variables show opposite small sample biases (see Nickell (1981)). We also find qualitatively similar estimates using quarterly cross-sectional Fama-MacBeth (1973) regressions and Newey-West (1987) standard errors for the coefficients. Including yearly time dummies in the pooled OLS regressions also does not affect our results.

Table I
Predicting Earnings Using Negative Words

This table shows estimates of the ability of negative words ($neg_{-30,-3}$) to predict quarterly earnings (SUE or $SAFE$) using ordinary least squares (OLS) regressions. We display the regression coefficients and summary statistics from six regressions below: two different dependent variables (SUE and $SAFE$) regressed on negative words computed based on four sets of news stories (*Dow Jones News Service (DJNS)*, *The Wall Street Journal (WSJ)*, “Before Forecasts,” and “All Stories”). SUE is a firm’s standardized unexpected quarterly earnings; and $SAFE$ is the standardized analysts’ forecast error for the firm’s quarterly earnings. The negative words variable ($neg_{-30,-3}$) is the standardized number of negative words in the news stories from 30 to 3 trading days prior to an earnings announcement divided by the total number of words in these news stories. The *DJNS* and *WSJ* regressions use only stories from these sources to compute $neg_{-30,-3}$. The two “Before Forecasts” regressions compute $neg_{-30,-3}$ using only stories that occur 1 trading day before the most recent consensus analyst forecast. All regressions include control variables for lagged firm earnings, firm size, book-to-market, trading volume, recent and distant past stock returns, and analysts’ quarterly forecast revisions and dispersion (see text for details). Following Froot (1989), we compute clustered standard errors by calendar quarter. The robust t -statistics are in parentheses.

Stories Included	<i>SUE</i>				<i>SAFE</i>	
	<i>DJNS</i>	<i>WSJ</i>	Before Forecasts	All Stories	Before Forecasts	All Stories
$neg_{-30,-3}$	-0.0584 (-4.42)	-0.1083 (-5.28)	-0.0640 (-3.95)	-0.0637 (-4.69)	-0.0192 (-3.79)	-0.0197 (-4.44)
<i>Lag(Dependent Var)</i>	0.2089 (11.82)	0.2082 (11.83)	0.2042 (11.90)	0.2101 (11.98)	0.2399 (7.82)	0.2523 (8.74)
<i>Forecast Dispersion</i>	-0.9567 (-9.84)	-1.0299 (-9.59)	-0.9634 (-9.21)	-0.9373 (-10.20)	-0.2984 (-5.34)	-0.3076 (-6.34)
<i>Forecast Revisions</i>	20.2385 (8.89)	18.0394 (7.91)	20.4855 (8.51)	19.5198 (8.94)	0.5111 (0.68)	0.7580 (1.19)
<i>Log(Market Equity)</i>	-0.0071 (-0.40)	0.0003 (0.01)	-0.0043 (-0.24)	-0.0037 (-0.21)	0.0258 (4.79)	0.0289 (5.32)
<i>Log(Book / Market)</i>	0.0173 (0.62)	0.0182 (0.56)	0.0221 (0.77)	0.0204 (0.75)	-0.0162 (-1.97)	-0.0110 (-1.41)
<i>Log(Share Turnover)</i>	-0.1241 (-3.09)	-0.1348 (-2.90)	-0.1095 (-2.75)	-0.1261 (-3.20)	0.0274 (2.69)	0.0254 (2.61)
$FFAlpha_{-252,-31}$	1.9784 (9.14)	1.9711 (9.90)	1.9770 (10.01)	2.0015 (9.50)	0.2199 (4.17)	0.2382 (4.36)
$FFCAR_{-30,-3}$	0.0119 (6.76)	0.0129 (6.33)	0.0117 (6.28)	0.0116 (6.64)	0.0062 (10.17)	0.0071 (11.38)
$FFCAR_{-2,-2}$	0.0104 (1.65)	0.0103 (1.37)	0.0177 (2.40)	0.0118 (1.91)	0.0053 (2.30)	0.0037 (1.89)
Observations	16755	11192	13722	17769	12907	16658
Clusters	80	79	78	80	78	79
Adjusted R^2	0.1177	0.1204	0.1158	0.1187	0.1163	0.1244

Table I display the coefficient estimates for all independent variables in these two regressions. As one would expect, several control variables exhibit strong explanatory power for future earnings. For example, lagged earnings, variables based on analysts’ forecasts, and recent stock returns ($FFCAR_{-30,-3}$) are all powerful predictors of earnings.

To gain intuition on the importance of language in predicting fundamentals, we compare the abilities of negative words in firm-specific news stories ($neg_{-30,-3}$) and firms' recent stock returns ($FFCAR_{-30,-3}$) to predict future earnings. The logic of this comparison is that both variables capture potentially relevant firm-specific information over the same time horizon—their correlation is -0.05 , and strongly statistically significant. This is a particularly challenging comparison for language because the firm's abnormal return measures the representative investor's interpretation of firm-specific news, which is undoubtedly based on a more sophisticated reading of the linguistic content that we quantify. In this respect, it is surprising that quantified language has any explanatory power above and beyond market returns. Indeed, one could view a firm's abnormal return ($FFCAR_{-30,-3}$) measured over the time horizon in which there is news ($[-30,-3]$) as an alternative quantification of the tone of news (e.g., Chan (2003)).

Surprisingly, Columns 4 and 6 in Table I reveal that negative words and recent stock returns have almost the same statistical impact and comparable economic impacts on future earnings. After standardizing the coefficients to adjust for the different variances of the independent variables, we find that the economic impact of past returns is 0.127 *SUE* and the impact of negative words is 0.063 *SUE*—roughly half as large. We infer that incorporating directly quantified language in earnings forecasts significantly improves upon using stock returns alone to quantify investors' reactions to news stories.

The “Before Forecasts” columns (3 and 5) in Table I show that negative words ($neg_{-30,-3}$) robustly predict both *SUE* and *SAFE* even after we exclude words from the most recent stories. Surprisingly, the respective $neg_{-30,-3}$ coefficients change in magnitude by less than 3% relative to Columns 4 and 6, and both remain strongly significant at any conventional level (p -values < 0.001).

In additional unreported tests, we run separate regressions for two subperiods, pre-1995 and 1995 to 2004, based on the idea that media coverage changed significantly in 1995 with the introduction of the Internet—for example, the *WSJ* officially launched *WSJ.com* on April 29, 1995. The main finding is that the significance and magnitude of all our results are quite similar for both subperiods. In summary, the evidence consistently shows that even a crude quantification of qualitative fundamentals ($neg_{-30,-3}$) can predict earnings above and beyond more recent measures of market prices and analysts' forecasts.

We now examine the long-run time-series behavior of earnings surrounding the release of negative words in firm-specific news stories. Figure 2 compares the earnings of firms with negative and positive news stories from 10 fiscal quarters prior to an earnings announcement up to 10 fiscal quarters after the earnings announcement. The dependent variable in Figure 2 is a firm's cumulative *SUE* beginning 10 quarters prior to the earnings announcement when the news was released. Our cumulative *SUE* computation does not discount earnings in different time periods. Using a positive discount rate would make the effect of negative words on earnings appear larger and more permanent.

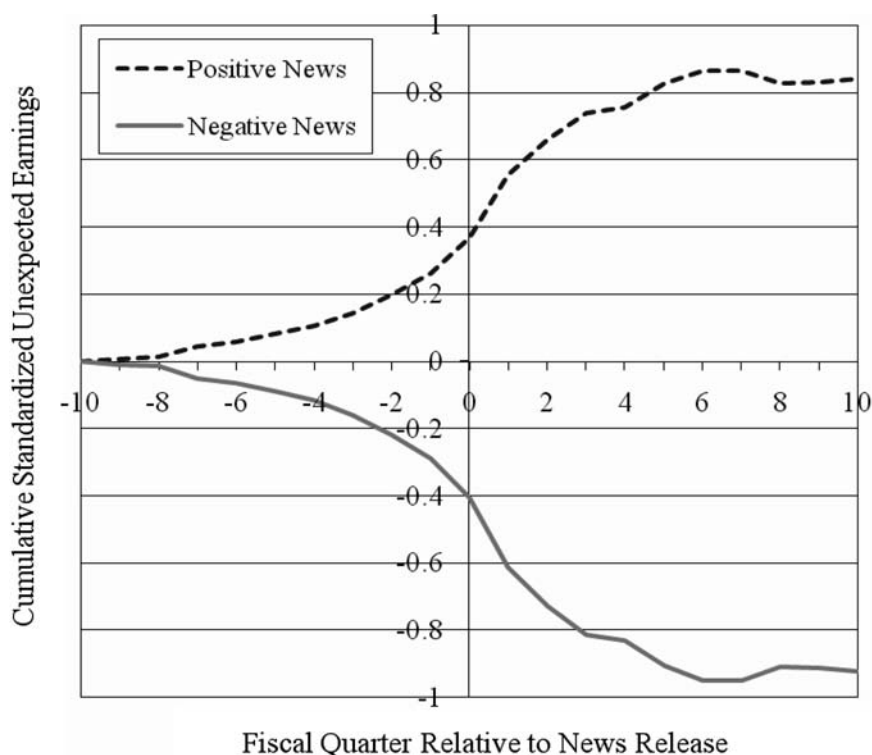


Figure 2. Firms' fundamentals around positive and negative news stories. In this figure, we graph firms' cumulative standardized unexpected earnings (*SUE*) from 10 fiscal quarters preceding media coverage of an earnings announcement to 10 quarters after the media coverage. We define media coverage of the announcement as positive (negative) when it contains a fraction of negative words ($Neg_{-30,-3}$) in the previous year's top (bottom) quartile. The measure of negative words ($Neg_{-30,-3}$) is the fraction of words that are negative in the news stories from 30 trading days prior up to 3 trading days prior to an earnings announcement. We separately analyze the firms with positive and negative media coverage prior to their earnings announcements. We compute the cumulative *SUE* for both sets of firms, beginning 10 quarters prior to the news and ending 10 quarters after the news. To compute *SUE* values after the news stories, we use only unexpected earnings benchmarks known at the time of the news—that is, those based on earnings prior to quarter zero (see text for details).

To compute *SUE* values after the news stories in Figure 2, we use only benchmarks for unexpected earnings that are known at the time of the news—that is, those based on earnings information prior to quarter zero. We use the matching seasonal earnings figure from before quarter zero to compute unexpected earnings after quarter zero—for example, we subtract E_{-3} from E_1 , E_5 , and E_9 to obtain UE_1 , UE_5 , and UE_9 . To obtain *SUE* measures, we standardize these unexpected earnings values using the mean and volatility of unexpected

earnings as measured in quarter zero.¹³ We define positive (negative) news as news in which the variable $Neg_{-30,-3}$ is in the bottom (top) quartile of the previous year's distribution of $Neg_{-30,-3}$.¹⁴

Figure 2 shows that firms with negative news stories before an earnings announcement experience large negative shocks to their earnings that endure for at least 4 quarters after the news. Although there are noticeable differences between firms with positive stories and those with many negative stories that appear before the news is released (0.772 cumulative *SUE*), the greatest discrepancy between the cumulative earnings of the two types of firms (1.816 cumulative *SUE*) appears in the sixth fiscal quarter after the news event. It appears as though most of the impact of negative words on cumulative earnings is permanent—1.764 cumulative *SUE* after 10 quarters, which is 0.992 cumulative *SUE* more than prior to the news. However, it is difficult to judge the magnitude and duration of the effect based on just 10 independent 10-quarter periods.

From the analysis above, we conclude that negative words in firm-specific stories leading up to earnings announcements significantly contribute to a useful measure of firms' fundamentals. One view is that this result is surprising because numerous stock analysts and investors closely monitor the actions of S&P 500 firms. Yet even after controlling for recent stock returns, analyst forecasts and revisions, and other measures of investors' knowledge, we find that a rudimentary linguistic measure of negative news still forecasts earnings. Furthermore, as we will demonstrate in Section V, it is possible to improve substantially upon this basic negative word count measure.

An alternative view is that negative words are informative measures of firms' fundamentals because they do not suffer from the same shortcomings as the quantitative variables that one can use to forecast earnings. For example, it is widely known that stock analysts' earnings forecasts exhibit significant biases that limit their forecasting power. In addition, stock market returns reflect revisions in investors' expectations of the present value of all future earnings as opposed to just next quarter's earnings, which is the dependent measure in our regressions. Even if investors and stock analysts are fully aware of the information embedded in negative words, negative words may have significant incremental explanatory power for future earnings because readily available quantitative variables are not accurate representations of investors' expectations.

¹³ We correct for the longer time intervals (T years) between the benchmark and unexpected earnings using the seasonal random walk assumption that the mean of unexpected earnings scales linearly (μT) and the volatility increases with the square root of the time interval ($\sigma T^{1/2}$). To mitigate any benchmarking biases, we also rescale *SUE* in each quarter so that its unconditional mean is zero, which affects the level of the lines in Figure 2 but has no impact on the difference between them.

¹⁴ As one would expect, the fractions of positive and negative words in news stories are negatively correlated (-0.18 , p -value < 0.001). For this reason, defining positive stories as those with relatively few negative words also produces stories with relatively more positive words.

IV. Using Negative Words to Predict Stock Returns

We subject the two competing views described above to empirical scrutiny in our return predictability tests. Having established that negative words in news stories can predict firms' fundamentals, we now examine whether they provide novel information not already represented in stock market prices. Unfortunately, we cannot test this conjecture by looking at contemporaneous market returns. Although there is a significant negative relationship between negative words and concurrent market returns, it is difficult to know which variable causes the other. Instead, we hypothesize that investors do not immediately respond in full to the news embedded in negative words. To test this theory, we explore whether negative words predict firms' future stock returns.

A. Predicting Returns in Story Event Time

In this subsection, we focus on OLS regression estimates of the effect of negative words on future stock returns in event time relative to the release of the news story. We use daily returns and news stories because these are the highest frequencies for which both data are reliably available—that is, all firms have daily returns for the entire sample, and the *WSJ* is a daily publication. Other benefits of this choice are that the news and return data frequency match each other and match the data frequency in Tetlock (2007).

Our main test assesses whether standardized fractions of negative words in firm-specific news stories on day zero predict firms' close-to-close stock returns on day one. For all *DJNS* stories, we obtain precise time stamp data to exclude stories that occur after 3:30 pm on day zero—that is, 30 minutes prior to market closing. To be conservative, we use the last time stamp for each story, which indicates when the story was most recently updated. Thus, in many cases, the negative words in *DJNS* stories became known to investors much earlier, often by 1 hour, than we assume. This ensures that traders have at least 30 minutes, and usually much longer, to digest and trade on the information in these stories. For all *WSJ* stories, we assume that stories printed in the morning's *WSJ* are available to traders well before the market close on the same day.

In each regression, we include several standard control variables to assess whether negative words predict returns above and beyond already-known sources of predictability, including both firms' characteristics (Daniel et al. (1997)) and firms' covariances with priced risk factors (Fama and French (1993)). We include all of the characteristic controls in the earnings predictability regressions, except the two analyst earnings forecast variables.¹⁵ That is, we include the firm's most recent earnings announcement (*SUE*), along with its

¹⁵ When we include the two analyst forecast variables, we find that both revisions and dispersion are statistically and economically insignificant predictors of returns in our sample. The coefficients on the key variables do not change materially. Thus, we omit the analyst variables to include any S&P 500 firms without analyst coverage and the first 4 years of our sample in the regression results.

close-to-close abnormal returns on the day of the news story ($FFCAR_{0,0}$), each of the previous 2 trading days ($FFCAR_{-1,-1}$ and $FFCAR_{-2,-2}$), the previous month ($FFCAR_{-30,-3}$), and the previous year ($FFAlpha_{-252,-31}$). These controls are designed to capture return predictability from past earnings (e.g., Ball and Brown (1968)) and past returns (e.g., Jegadeesh and Titman (1993)), which may be distinct phenomena (e.g., Chan, Jegadeesh, and Lakonishok (1996)). In addition, we control for firm size and book-to-market ratios using each firm's log of market capitalization and log of book-to-market equity measured at the end of the most recent June. These controls mimic the variables that Fama and French (1992) use to predict returns. We also control for trading volume using the log of share turnover.

We run two sets of regressions to ensure that firms' return covariances with priced risk factors do not drive our results. In the first set of regressions, we use each firm's next-day abnormal return as the dependent variable, where the Fama-French three-factor model is the benchmark for expected returns.¹⁶ To ensure that our results do not depend heavily on the benchmarking process, we run a second set of regressions in which we use each firm's next-day raw return as the dependent variable.

Table II reports the results from six OLS regressions, two different dependent variables (raw and abnormal next-day returns) regressed on each of three different negative words measures (*DJNS*, *WSJ*, and "All Stories"). The table shows the coefficients on negative words in firm-specific news stories and their associated *t*-statistics. We compute clustered standard errors (Froot (1989)) to account for the correlations between firms' stock returns within trading days. The reasons for choosing OLS regression estimates and standard errors clustered by time period are analogous to those discussed in Section III and are not repeated here. Table II reports the number of clusters—that is, trading days—and the adjusted R^2 for each regression.

The main result in Table II is that negative words in firm-specific news stories robustly predict slightly lower returns on the following trading day. The coefficients on negative words (*neg*) are consistently significant in all four of the regressions where news stories from *DJNS* are included. The magnitude of the *DJNS* regression coefficient on *neg*, which is already standardized, implies that next-day abnormal returns ($FFCAR_{+1,+1}$) are 3.20 basis points lower after each one-standard deviation increase in negative words.

Interestingly, the coefficients on negative words are less than half as large and statistically insignificant in the two regressions where only *WSJ* stories are included. One interpretation is that *DJNS* releases intraday stories with extremely recent information before the information is fully priced. By contrast, many morning *WSJ* stories are recapitulations of the previous day's events—some of which appeared in the *DJNS*—that may already be incorporated in market prices.

¹⁶ We also find that including time dummies for each trading day—that is, demeaning returns by trading day—does not change our results, suggesting an omitted common news factor is not driving our results.

Table II
Predicting Returns Using Negative Words

This table shows the relationship between standardized fractions of negative words (*neg*) in firm-specific news stories and firms' stock returns on the following day ($Return_{+1,+1}$ or $FFCAR_{+1,+1}$). The coefficients on $neg_{-30,-3}$ and summary statistics from six regressions are displayed below: two different dependent variables ($Return_{+1,+1}$ and $FFCAR_{+1,+1}$) regressed on negative words from each of three sets of news stories (*Dow Jones News Service [DJNS]*, *The Wall Street Journal [WSJ]*, and all stories). We exclude stories that occur after 3:30 pm (30 minutes prior to market closing). We assume that *WSJ* stories printed in the morning's *WSJ* are available to traders before the market close on the same day. The two dependent variables are the firm's raw close-to-close return ($Return_{+1,+1}$) and the firm's abnormal return ($FFCAR_{+1,+1}$). We use the Fama-French three-factor model with a $[-252, -31]$ trading day estimation period relative to the release of the news story as the benchmark for expected returns. The key independent variable is *neg*, the fraction of negative words in each news story standardized using the prior year's distribution. Each regression also includes control variables for the firm's most recent earnings announcement (*SUE*), market equity, book-to-market equity, trading volume, and close-to-close returns on the day of the news story, each of the previous two trading days, and the previous calendar year. Following Froot (1989), we compute clustered standard errors by trading day. The robust *t*-statistics are in parentheses.

Stories Included	$Return_{+1,+1}$			$FFCAR_{+1,+1}$		
	<i>DJNS</i>	<i>WSJ</i>	All	<i>DJNS</i>	<i>WSJ</i>	All
<i>neg</i>	-0.0277 (-3.67)	-0.0105 (-1.24)	-0.0221 (-3.72)	-0.0320 (-4.83)	-0.0102 (-1.37)	-0.0253 (-4.88)
$FFCAR_{0,0}$	0.0285 (5.28)	0.0229 (2.92)	0.0246 (5.43)	0.0259 (5.00)	0.0224 (2.94)	0.0226 (5.19)
$FFCAR_{-1,-1}$	-0.0272 (-3.63)	-0.0154 (-2.17)	-0.0222 (-4.21)	-0.0254 (-3.86)	-0.0106 (-1.68)	-0.0190 (-4.13)
$FFCAR_{-2,-2}$	-0.0215 (-3.16)	-0.0094 (-1.10)	-0.0179 (-3.39)	-0.0207 (-3.10)	-0.0104 (-1.22)	-0.0183 (-3.60)
$FFCAR_{-30,-3}$	-0.0005 (-0.30)	0.0016 (0.73)	-0.0002 (-0.13)	0.0004 (0.28)	0.0018 (0.85)	0.0005 (0.38)
$FFAlpha_{-252,-31}$	0.0559 (0.57)	0.1470 (1.29)	0.1046 (1.27)	0.1201 (1.36)	0.1686 (1.67)	0.1465 (2.02)
Earnings (<i>SUE</i>)	0.0160 (2.84)	0.0082 (1.33)	0.0125 (2.68)	0.0152 (3.46)	0.0055 (1.09)	0.0115 (3.25)
$\text{Log}(\text{Market Equity})$	-0.0152 (-2.02)	-0.0159 (-1.99)	-0.0154 (-2.39)	-0.0120 (-2.19)	-0.0121 (-1.97)	-0.0109 (-2.51)
$\text{Log}(\text{Book / Market})$	-0.0027 (-0.18)	0.0087 (0.60)	-0.0010 (-0.08)	-0.0246 (-2.12)	-0.0061 (-0.52)	-0.0201 (-2.22)
$\text{Log}(\text{Share Turnover})$	-0.0324 (-1.66)	-0.0278 (-1.43)	-0.0300 (-1.76)	-0.0189 (-1.35)	-0.0167 (-1.16)	-0.0145 (-1.27)
Observations	141541	84019	208898	141541	84019	208898
Clusters (Days)	6260	6229	6272	6260	6229	6272
Adjusted R^2	0.0024	0.0014	0.0018	0.0026	0.0014	0.0019

We now analyze the $Return_{+1,+1}$ and $FFCAR_{+1,+1}$ regressions that include stories from the *DJNS* (in Columns 2 and 5 of Table II) in greater detail. As one would expect in an efficient market, very few control variables predict next-day returns, which is why the R^2 statistics in Table II are so low. Aside from the daily news and returns variables, only firms' earnings (*SUE*) have predictive power at the 1% level.

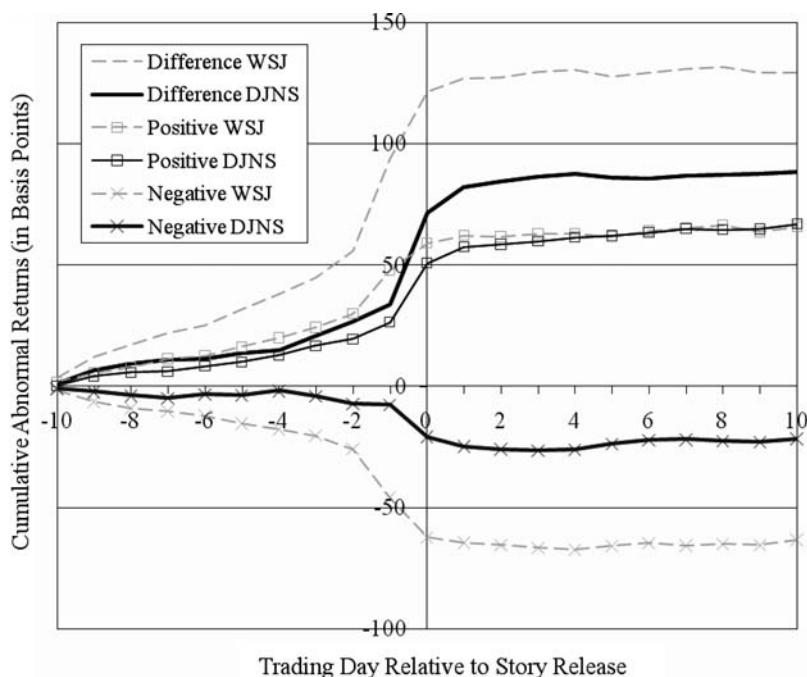


Figure 3. Firms' valuations around positive and negative news stories. In this figure, we graph a firm's abnormal event returns from 10 trading days preceding a news story's release to 10 trading days following its release. All news stories focus on S&P 500 firms and come from either *Dow Jones News Service* or *The Wall Street Journal* between 1980 and 2004 inclusive. For all *DJNS* stories, we exclude stories that occur after 3:30 pm (30 minutes prior to market closing). For all *WSJ* stories, we assume that stories printed in the morning's *WSJ* are available to traders well before the market close on the same day. We use the Fama-French three-factor model with a $[-252, -31]$ trading day estimation period relative to the release of the news story as the benchmark for expected returns. We label all news stories with a fraction of negative words (*Neg*) in the previous year's top (bottom) quartile as negative (positive) stories. We separately examine the market's response to positive and negative *DJNS* and *WSJ* stories. We also compute the difference between the reaction to positive and negative news stories for each source.

One pattern in these regressions is somewhat analogous to the main result in Chan (2003). He shows that stocks in the news experience annual return continuations, whereas those not in the news experience annual return reversals. Although Table II examines daily horizons, the interpretation of the day 0 (day-of-news), and day -1 and -2 (usually not news days) returns coefficients is quite similar. The positive coefficient on $FFCAR_{0,0}$ shows that news-day returns continue on the next day, whereas the negative coefficients on $FFCAR_{-1,-1}$ and $FFCAR_{-2,-2}$ show that non-news-day returns reverse themselves.

We now examine the market's apparently sluggish reaction to negative words in the 4 weeks surrounding the story's release to the public. Figure 3 graphs a firm's abnormal returns from 10 trading days before a story's release to 10 trading days after its release. Again, we use the Fama-French three-factor model to

estimate abnormal returns. We label all news stories with a fraction of negative words (*Neg*) in the previous year's top (bottom) quartile as negative (positive) stories. We separately examine the market's response to positive and negative *DJNS* and *WSJ* stories. We also compute the difference between the reaction to positive and negative news stories for each source.

Although Figure 3 shows that the market reacts quite efficiently to positive and negative news, there is some delayed reaction, particularly for the *DJNS* news stories. From the top line in Figure 3, one can see that the 12-day market reaction, from day -2 to day 10, to *WSJ* stories is virtually complete after the first 2 trading days—7.5 basis points (bps) of underreaction after day 1 and only 2.4 bps after day 2. By contrast, the second line in Figure 3 shows that more of the 12-day market reaction to *DJNS* stories persists beyond the first 2 days—16.8 bps after day 1 and 6.2 bps after day 2.

The *DJNS* lines in Figure 3 show the day 1 delayed reaction to positive *DJNS* news stories (6.6 bps) is somewhat larger than the delayed reaction to negative stories (4.0 bps).¹⁷ Although the total day 1 delayed reaction to *DJNS* news stories is 10.6 bps (see the difference line), this magnitude is relatively small (17.2%) as a percentage of the total 12-day reaction of roughly 61.6 bps. The market appears even more efficient in its reaction to *WSJ* stories, where the 1-day delayed reaction (5.2 bps) is only 7.1% of the 12-day reaction (73.3 bps). However, there may be additional underreaction to *WSJ* stories within the trading day that encompasses the morning release of the newspaper.

B. Predicting Returns in Calendar Time

The lingering difference between the abnormal returns of firms with positive and negative *DJNS* news stories suggests that a simple trading strategy could earn positive risk-adjusted profits. In this section, we explore this possibility, focusing on the apparent short-run underreaction to negative words in the *DJNS*.

Specifically, at the close of each trading day, we form two equal-weighted portfolios based on the content of each firm's *DJNS* news stories during the prior trading day.¹⁸ We use the same definitions for positive and negative stories as before. We include all firms with positive *DJNS* news stories from 12:00 am to 3:30 pm on the prior trading day in the long portfolio, and put all firms with negative stories in the short portfolio. We hold both the long and short portfolios for 1 full trading day and rebalance at the end of the next trading day. To keep the strategy simple, we exclude the rare days in which either the long or the short portfolio contains no qualifying firms. Ignoring trading costs, the cumulative raw returns of this long-short strategy would be 21.1% per year.

¹⁷ The contemporaneous reactions to positive news stories are also larger. We observe the opposite asymmetry for the positive and negative news stories about fundamentals that we examine in Section IV.

¹⁸ Forming two story-weighted or value-weighted portfolios produces very similar results.

Table III
Risk-Adjusted News-Based Trading Strategy Returns

This table shows the daily risk-adjusted returns (*Alpha*) from a news-based trading strategy for three different time periods (1980 to 1994, 1995 to 2004, and 1980 to 2004). The first three regressions use the Fama-French (1993) three-factor model to adjust the trading strategy returns for the impact of contemporaneous market (*Market*), size (*SMB*), and book-to-market (*HML*) factors. The last three regressions use the Carhart (1997) four-factor model to account for incremental impact of the momentum factor (*UMD*). Table III reports the alpha and loadings from the time-series regression of the long-short news-based portfolio returns on each of the four factors. We assemble the portfolio for the trading strategy at the close of each trading day. We form two equal-weighted portfolios based on the content of each firm's *Dow Jones News Service* stories during the prior trading day. We label all news stories with a fraction of negative words in the previous year's top (bottom) quartile as negative (positive) stories. We include all firms with positive news stories in the long portfolio and all firms with negative news stories in the short portfolio. We hold both the long and short portfolios for one full trading day and rebalance at the end of the next trading day. We exclude the rare days in which there are no qualifying firms in either the long or the short portfolio. We compute all coefficient standard errors using the White (1980) heteroskedasticity-consistent covariance matrix. The robust *t*-statistics are in parentheses.

	1980–1994	1995–2004	1980–2004	1980–1994	1995–2004	1980–2004
<i>Alpha</i>	0.0919 (2.83)	0.1175 (3.93)	0.1031 (4.55)	0.0952 (2.81)	0.1131 (3.78)	0.1013 (4.38)
<i>Market</i>	–0.0994 (–0.93)	–0.1087 (–1.99)	–0.0983 (–1.86)	–0.0831 (–0.75)	–0.1001 (–1.87)	–0.0999 (–1.87)
<i>SMB</i>	–0.0767 (–0.35)	0.0475 (0.70)	–0.0081 (–0.08)	–0.0647 (–0.29)	0.0341 (0.49)	–0.0128 (–0.12)
<i>HML</i>	–0.1869 (–1.24)	–0.2590 (–2.81)	–0.2372 (–2.94)	–0.1819 (–1.20)	–0.2500 (–2.75)	–0.2365 (–2.93)
<i>UMD</i>				–0.0911 (–0.74)	0.0930 (2.01)	0.0444 (0.90)
Trading Days	3398	2497	5895	3398	2497	5895
Adjusted R^2	0.0003	0.0081	0.0026	0.0004	0.0106	0.0027

Table III shows the risk-adjusted daily returns from this daily news-based trading strategy for three different time periods (1980 to 1994, 1995 to 2004, and 1980 to 2004). We use the Fama-French three-factor (1993) and Carhart four-factor (1997) models to adjust the trading strategy returns for the returns of contemporaneous market, size, book-to-market, and momentum factors. Table III reports the alpha and factor loadings from the time series regression of the long-short news-based portfolio returns on the four factors. The first three columns report the results with the Fama-French benchmark, whereas the last three columns use the Carhart benchmark. We compute all coefficient standard errors using the White (1980) heteroskedasticity-consistent covariance matrix.

Consistent with Table II, Table III shows that the daily news-based trading strategy would earn substantial risk-adjusted returns in a frictionless world with no trading costs or price impact. Specifically, the average excess return (Fama-French alpha) from news-based trading would be 9.2 bps per day from 1980 to 1994 and 11.8 bps per day from 1995 to 2004. Using any return benchmark, the alpha from the trading strategy is highly significant in all three

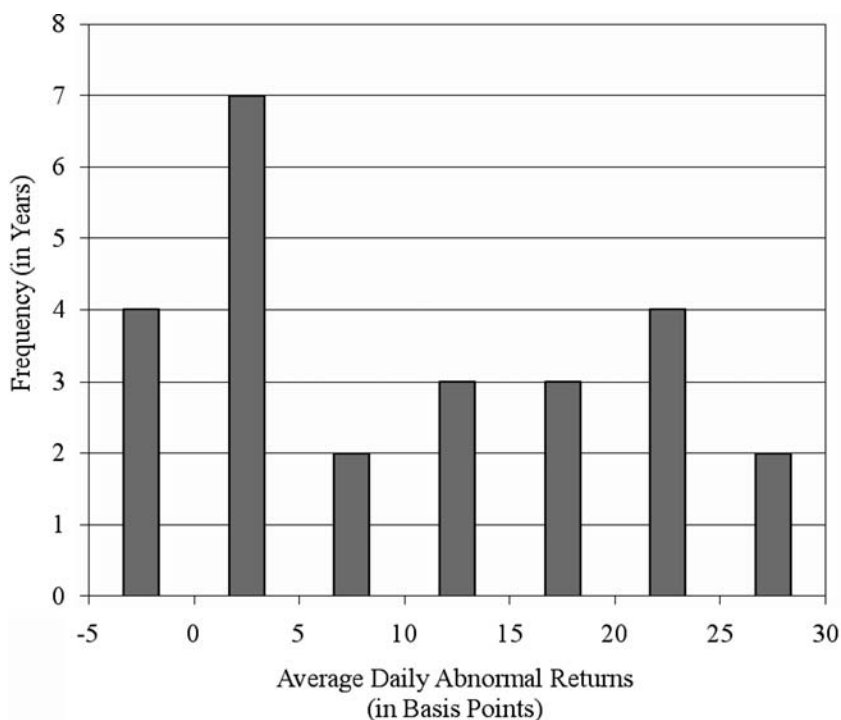


Figure 4. Distribution of daily abnormal returns for the news-based trading strategy.

For the 25 years from 1980 to 2004, the figure depicts the distribution of the average daily abnormal returns for the news-based trading strategy described below. Each frequency bin encompasses a 5% range of abnormal returns described by the two numbers adjacent to the bin—for example, the frequency of the leftmost return bin (4 years) is the number of years in which the trading strategy's average daily return is between -5 basis points and zero basis points. We assemble the portfolio for the news-based trading strategy at the close of each trading day. We form two equal-weighted portfolios based on the content of each firm's *Dow Jones News Service* stories during the prior trading day. We label all news stories with a fraction of negative words in the previous year's top (bottom) quartile as negative (positive) stories. We include all firms with positive news stories in the long portfolio and all firms with negative news stories in the short portfolio. We hold both the long and short portfolios for 1 full trading day and rebalance at the end of the next trading day. We exclude the rare days in which either the long or the short portfolio contains no qualifying firms. To adjust the returns for risk, we use the full-sample estimates of the Fama-French three-factor loadings of the news-based portfolio displayed in Table III.

time-periods. Interestingly, the returns from news-based trading are not strongly related to any of the Fama-French factors or the momentum factor.¹⁹ The very low R^2 statistics show that nearly all of the trading strategy risk is firm-specific, as one might expect because we focus on firm-specific news stories.

For the 25 years between 1980 and 2004, Figure 4 depicts the distribution of the average daily abnormal returns for the news-based trading strategy. In the

¹⁹ The strategy's negative loading on HML is a minor exception, possibly driven by the numerous positive news stories about growth firms during the late 1990s.

Table IV
Sensitivity of News-Based Trading Returns
to Trading Cost Assumptions

This table shows estimates of the impact of transaction costs on the news-based trading strategy's profitability (see the text or Table III for strategy details). We recalculate the trading strategy returns for 11 alternative assumptions about a trader's round-trip transaction costs: 0, 1, 2, 3 . . . or 10 basis points (bps) per round-trip trade. The abnormal and raw annualized cumulative news-based strategy returns for each assumption appear below. The risk-adjustment is based on the full-sample Fama-French three-factor loadings of the news-based portfolio shown in Table III.

Trading Costs (bps)	Abnormal Annualized Returns (%)	Raw Annualized Returns (%)
0	23.17	21.07
1	20.30	18.25
2	17.50	15.49
3	14.76	12.80
4	12.09	10.17
5	9.47	7.60
6	6.92	5.09
7	4.43	2.64
8	1.99	0.25
9	-0.39	-2.09
10	-2.71	-4.37

median year, the strategy's abnormal return is 9.4 bps per day. In 21 out of 25 years, the news-based strategy earns positive abnormal returns. Thus, we can reject the null hypothesis that yearly news-based strategy returns follow the binomial distribution with an equal likelihood of positive and negative returns (p -value < 0.0005). There is only 1 year (1980) out of 25 in which the strategy lost more than 2 bps per day (-4.2 bps). By contrast, in 6 out of 25 years, the strategy gained more than 20 bps per day. This analysis suggests that the news-based trading strategy is not susceptible to catastrophic risks that second moments of returns may fail to capture.

Finally, we estimate the impact of reasonable transaction costs on the trading strategy's profitability. To judge the sensitivity of profits to trading costs, we recalculate the trading strategy returns under the assumption that a trader must incur a round-trip transaction cost of between zero and 10 bps. Table IV displays the abnormal and raw annualized cumulative news-based strategy returns under these cost assumptions.

From the evidence in Table IV, we see that the simple news-based trading strategy explored here is no longer profitable after accounting for reasonable levels of transaction costs—for example, 10 bps. Of course, we cannot rule out the possibility that more sophisticated trading rules that exploit the time-series and cross-sectional properties of news stories and economize on trading costs would be profitable. For example, the next subsection investigates a refined measure of negative words that predicts greater market underreactions to particular negative words.

V. Interpreting the Earnings and Return Predictability

The key stylized facts documented thus far are: (1) news stories about firms are concentrated around their earnings announcements; (2) negative words in firm-specific stories predict low firm earnings in the next quarter; and (3) negative words about firms predict low firm stock returns on the next trading day. In this section, we explore further whether the ability of negative words to predict returns comes from underreaction to news about firms' fundamentals that is embedded in language.

Our specific hypothesis is that negative words in news stories that mention the word stem "earn" contain more information about firms' fundamentals than other stories. If this is the case, we should observe three effects. First, negative words in stories that include the word stem "earn" should be better predictors of earnings. Second, the contemporaneous relationship between negative words and returns should be stronger for stories that contain the word stem "earn." Third, because these stories better capture news about hard-to-quantify fundamentals, the magnitude of the market's underreaction to negative words should be greater for stories that contain the word stem "earn."²⁰

Before testing these three predictions, we establish an intuitive property of this measure of fundamentals: the news stories near earnings announcements (see the spike in Figure 1) are far more likely to mention the word stem "earn"—for example, the word "earnings" or any form of the verb "earn." We construct a dummy variable (*Fund*) that indicates whether a news story contains any words beginning with "earn." We find that only 18.9% of the stories more than 1 day away from an earnings announcement contain the word stem "earn," compared to 72.5% of the stories within a day of an announcement.

We test whether negative words in stories containing the word stem "earn" predict earnings better than negative words in other stories. We add two new independent variables to the regressions for *SUE* and *SAFE* shown earlier in Columns 4 and 6 of Table I. The first new variable ($Fund_{-30,-3}$) is the total number of words in news stories between day -30 and day -3 that contain the word stem "earn" divided by the total number of words in all news stories between day -30 and day -3 . This measure is designed to capture the fraction of words between day -30 and day -3 that are likely to provide relevant information about firms' fundamentals. The second new variable ($neg_{-30,-3} * Fund_{-30,-3}$) is the interaction between $Fund_{-30,-3}$ and the negative words measure ($neg_{-30,-3}$). The coefficient on the interaction term measures the extent to which negative words "about" fundamentals are more useful predictors of firms' earnings than other negative words.

Table V shows that the coefficients for both of the new independent variables in the *SUE* and *SAFE* regressions are strongly negative and statistically significant. The coefficient on the interaction term ($neg_{-30,-3} * Fund_{-30,-3}$) in the *SUE* regression shows that negative words that are "about" fundamentals are

²⁰ Pritamani and Singhal (2001) document a fact that may be related to this third hypothesis. Although they do not examine the tone of news stories, they do find return momentum following market reactions to earnings-related news stories.

Table V

Predicting Earnings Using Negative Words about Fundamentals

This table reports the results from two OLS regressions with different dependent variables (*SUE* and *SAFE*) regressed on negative words ($neg_{-30,-3}$), fundamental words ($Fund_{-30,-3}$), and the interaction between these words ($neg_{-30,-3} * Fund_{-30,-3}$). *SUE* is a firm's standardized unexpected quarterly earnings; *SAFE* is the standardized analysts' forecast error for the firm's quarterly earnings. Both regressions include all news stories from both news sources (*Dow Jones News Service* and *The Wall Street Journal*) over the time period from 1984 through 2004. The measure of negative words ($neg_{-30,-3}$) is the standardized fraction of words that are negative in the news stories from 30 trading days prior up to 3 trading days prior to an earnings announcement. Fundamental words ($Fund_{-30,-3}$) is the fraction of words that are contained in news stories that mention the word stem "earn" from 30 trading days prior up to 3 trading days prior to an earnings announcement. All regressions include control variables for lagged firm earnings and numerous firm characteristics (see text for details). To allow for correlations among announced firm earnings within the same calendar quarter, we compute clustered standard errors (Froot (1989)). The robust *t*-statistics are in parentheses.

	<i>SUE</i>	<i>SAFE</i>
$neg_{-30,-3}$	-0.0167 (-1.19)	-0.0072 (-1.65)
$neg_{-30,-3} * Fund_{-30,-3}$	-0.3192 (-8.00)	-0.0824 (-5.48)
$Fund_{-30,-3}$	-0.4676 (-7.27)	-0.1033 (-5.59)
<i>Lag(Dependent Var)</i>	0.2080 (12.22)	0.2517 (8.69)
<i>Forecast Dispersion</i>	-0.9280 (-10.32)	-0.3049 (-6.35)
<i>Forecast Revisions</i>	19.1856 (9.06)	0.7068 (1.11)
<i>Log(Market Equity)</i>	-0.0062 (-0.36)	0.0285 (5.31)
<i>Log(Book / Market)</i>	0.0126 (0.48)	-0.0127 (-1.60)
<i>Log(Share Turnover)</i>	-0.1086 (-2.89)	0.0299 (3.08)
$FFAlpha_{-252,-31}$	1.9760 (9.59)	0.2317 (4.30)
$FFCAR_{-30,-3}$	0.0102 (5.74)	0.0067 (11.20)
$FFCAR_{-2,-2}$	0.0110 (1.81)	0.0036 (1.87)
Observations	17769	16658
Clusters	80	79
Adjusted R^2	0.1282	0.1285

much better predictors of firms' earnings. Because the $Fund_{-30,-3}$ variable is a fraction that ranges from zero to one, the regression coefficients have meaningful economic interpretations. The sum of the coefficient on the interaction ($neg_{-30,-3} * Fund_{-30,-3}$) and the coefficient on negative words alone ($neg_{-30,-3}$)

estimates the dependence of firm earnings on negative words for announcements in which all ($Fund_{-30,-3} = 1$) of the news stories between day -30 and day -3 contain the stem “earn.” The coefficient on negative words ($neg_{-30,-3}$) now estimates the dependence of firm earnings on negative words when none ($Fund_{-30,-3} = 0$) of the news stories between day -30 and day -3 contain the stem “earn.” Also, one can recover the direct effect of negative words in a typical set of news stories, where 26.3% of the words are about earnings ($Fund_{-30,-3} = 0.263$), by computing (coefficient on $neg_{-30,-3}$) + $0.263 \times$ (coefficient on $neg_{-30,-3} * Fund_{-30,-3}$). This last quantity is directly comparable to the simple coefficients on $neg_{-30,-3}$ that appear in Table I.

The point estimate of the sum of the interaction term and the $neg_{-30,-3}$ coefficient ($-0.3359 SUE$) is over 10 times greater than the $neg_{-30,-3}$ coefficient ($-0.0167 SUE$), suggesting that negative words derive almost all of their predictive power for SUE from earnings-related stories. Negative words in stories unrelated to earnings (see coefficients on $neg_{-30,-3}$) only weakly predict lower earnings, and are much less important in economic terms. Yet the direct effect of negative words on earnings in a “typical” set of stories with 26.3% earnings-related words remains strongly statistically and economically significant at $-0.0167 + 0.263 \times (-0.3192) = -0.1006 SUE$. Similarly, negative words in earnings-related stories can predict analyst forecast errors ($SAFE$) better by an order of magnitude.

We now test the other two predictions of our hypothesis: Contemporaneous market reactions and subsequent market underreactions should be larger for stories that mention the word stem “earn” than for other stories. As before, we use pooled OLS regressions with clustered standard errors to estimate the relationship between negative words and returns. We also use the same set of firm characteristic and stock return control variables. To conserve space, we report only the results where we use firms’ abnormal returns as the dependent variable and negative words in firm-specific stories from *DJNS* as the key independent variable. Again, we use the *DJNS* stories that occur more than 30 minutes before the market closes to explore the underreaction hypothesis because Table II reveals that there is only minimal underreaction to *WSJ* stories.

Column 1 in Table VI reports the contemporaneous (same-day) relationship between abnormal returns ($FFCAR_{+0,+0}$) and negative words (neg). There are two new independent variables in these regressions: the dummy variable that is equal to one if a story mentions the word stem “earn” ($Fund$) and the interaction ($neg * Fund$) between this dummy variable and standardized negative words (neg).

Column 1 in Table VI reveals that not only is there a strong relationship between negative words (neg) and contemporaneous returns ($FFCAR_{+0,+0}$), but also that this relationship is much stronger in stories that contain the “earn” word stem dummy. The sum of the coefficient on neg by itself (-8.57 bps) and the coefficient on the $neg * Fund$ term (-31.27 bps) provides an estimate of the contemporaneous market response to negative words in news stories that mention earnings (-39.84 bps). In economic terms, the coefficient magnitudes

Table VI
Firms' Returns and Negative Words about Fundamentals

This table shows the relationship between negative words in firm-specific news stories (*neg*) and firms' close-to-close abnormal stock returns on the same day ($FFCAR_{+0,+0}$) and the following day ($FFCAR_{+1,+1}$). The stories include all *Dow Jones News Service* articles from 1980 through 2004, but exclude stories that occur after 3:30 pm (30 minutes prior to market closing). The coefficients and summary statistics from two OLS regressions using two different dependent variables ($FFCAR_{+0,+0}$ and $FFCAR_{+1,+1}$) appear below. The key independent variable is negative words (*neg*), which is the fraction of negative words in each news story standardized using the prior year's distribution. The independent variable *Fund* is a dummy indicating whether a story mentions the word stem "earn"; and *neg * Fund* is the interaction between negative words (*neg*) and this dummy. All regressions include numerous control variables for lagged firm returns and other firm characteristics (see text for details). To allow for correlations among firms' returns within the same trading day, we compute clustered standard errors (Froot (1989)). The robust *t*-statistics are in parentheses.

	$FFCAR_{+0,+0}$	$FFCAR_{+1,+1}$
<i>neg</i>	-0.0857 (-11.16)	-0.0161 (-2.35)
<i>neg * Fund</i>	-0.3127 (-10.75)	-0.1036 (-4.52)
<i>Fund</i>	-0.3250 (-12.84)	-0.0342 (-1.96)
$FFCAR_{+0,+0}$		0.0255 (4.91)
$FFCAR_{-1,-1}$	0.0181 (2.23)	-0.0256 (-3.89)
$FFCAR_{-2,-2}$	-0.0220 (-2.72)	-0.0209 (-3.13)
$FFCAR_{-30,-3}$	0.0020 (1.17)	0.0004 (0.23)
$FFAlpha_{-252,-31}$	-0.1425 (-1.36)	0.1136 (1.29)
<i>Earnings (SUE)</i>	0.0234 (4.16)	0.0146 (3.33)
Observations	141633	141541
Clusters	6260	6260
Adjusted R^2	0.0045	0.0028

mean that the market response to negative words in earnings-related stories is five times larger than the response to other negative words. This evidence supports the hypothesis that negative words convey otherwise hard-to-quantify information about firms' fundamentals.²¹

In Column 2 of Table VI, we repeat the previous regression, except we use firms' next-day abnormal returns ($FFCAR_{+1,+1}$) as the dependent variable. The main result is that the same variables that elicit the greatest contemporaneous market responses also predict the greatest subsequent market underreaction.

²¹ The negative market responses to the presence of earnings-related words (*Fund*) could represent earnings warnings from firm management prior to earnings announcements.

For example, the coefficient on the interaction term ($neg * Fund$) is highly negative (Column 2 in Table VI), showing that negative words in earnings-related stories predict greater market underreactions than negative words in other stories (neg). In fact, the market's underreaction to negative words in stories not mentioning earnings is only one-seventh as large as its underreaction to negative words in stories about earnings (-1.61 bps vs. -11.97 bps $= -1.61 - 10.36$). The estimate of underreaction to negative words in "typical" stories that mention earnings with probability 28.0% lies between the two previous values at -4.33 bps $= -1.61$ bps $+ 0.280 \times (-10.36)$ bps, and remains highly statistically significant.

We can gauge the degree of underreaction by comparing the sizes of the 1-day and 2-day reactions to negative words. Table VI allows us to make this comparison for negative words in earnings-related stories and those in other stories. The coefficients on neg in the first column (-8.57 bps) and second column (-1.61 bps) measure the day 0 and day 1 reactions for negative words unrelated to earnings. The sums of the coefficients on neg and $neg * Fund$ in the first column (-39.84 bps) and second column (-11.97 bps) measure the day 0 and day 1 reactions for earnings-related words. From this, we can infer that the market's initial 1-day reaction to negative words comprises the vast majority of its 2-day reaction for stories unrelated (84.2%) and related (76.9%) to earnings. One interpretation is that investors remain almost equally attuned to the importance of linguistic information about fundamentals even during earnings announcements, when there is compelling quantitative information.

All three tests in this section suggest that negative words in stories about firms' fundamentals are driving the earnings and return predictability results. Although news stories that do not mention earnings can weakly forecast earnings and are associated with contemporaneous market returns, these stories have very little ability to forecast future market returns. Negative words in earnings-related stories evoke much greater initial market responses presumably because these stories are better predictors of firms' subsequent earnings. However, the initial market responses to negative words in earnings-related stories are insufficiently large to prevent return continuations on the next trading day. Investors seem to distinguish between earnings-related stories and others, but do not fully account for the importance of linguistic information about fundamentals.

VI. Conclusion

Our first main result is that negative words in the financial press forecast low firm earnings. That is, the words contained in news stories are not redundant information, but instead capture otherwise hard-to-quantify aspects of firms' fundamentals. Our second result is that stock market prices incorporate the information embedded in negative words with a slight delay. We demonstrate potential profits from using a simple trading strategy based on the words in a timely news source (*DJNS*), but find that these profits could easily vanish after accounting for reasonable levels of transaction costs. Finally, we show that

negative words in stories about fundamentals are particularly useful predictors of both earnings and returns.

Our overall impression is that the stock market is relatively efficient with respect to firms' hard-to-quantify fundamentals. The market's underreaction to negative words after day 0 is typically small as compared to the market's initial reaction to negative words on day 0. Even if economists have neglected the possibility of quantifying language to measure firms' fundamentals, stock market investors have not.

Nevertheless, we do find that market prices consistently underreact to negative words in firm-specific news stories, especially those that relate to fundamentals. Although frictionless asset pricing models may not be able to explain these findings, models in which equilibrium prices induce traders to acquire costly information—for example, Grossman and Stiglitz (1980)—are broadly consistent with our results. Without some slight underreaction in market prices, traders would have no motivation to monitor and read the daily newswires. Future research on quantifying language has the potential to improve our understanding of how information is incorporated in asset prices.

Appendix

To match firms' names in CRSP with their common names used in the media, we employ a combination of four methods. Our first method works well for firms that are currently members of the S&P 500 index. We download common names for these firms from the "S&P constituents" spreadsheet posted on Standard and Poor's Web site, <http://www.standardpoor.com/>. We match these common names to CRSP name strings, which we use in our Factiva news queries for the 473 firms in the S&P at the end of our data period (December 31, 2004) that remained in the index on the date that we downloaded the spreadsheet. We identify the common names of the other 27 S&P 500 firms at the end of 2004 using the methods described below.

The other three methods entail matching the CRSP name strings with common firm names from one of three Web-based data sources: Mergent Online, the Securities and Exchange Commission (SEC), or Factiva. For all companies that exist after 1993, we use the Mergent Online company search function to identify firms' common names (336 firms). For the few post-1993 companies without Mergent data, we use the SEC company name search function (20 firms). Finally, we identify the common names of firms prior to 1993 using the Factiva company name search function (285 firms).

In many cases, we manually tweak the CRSP names to improve the quality of the company search. For example, if we do a company search for the CRSP name string "PAN AMERN WORLD AWYS INC," Factiva returns no results. Logically, we look for "Pan American," which seems to retrieve the appropriate company name: "Pan American World Airways Inc." Although this matching process introduces the possibility of minor judgment errors, our searches uniquely identify matching firms in all cases, suggesting our methods are reasonable.

We construct search queries for news stories using the common names that we match to the CRSP name string. We spot-check all stories that mention S&P 500 firms in the *DJNS* and *WSJ* to ensure that our search criteria do not exclude too many stories that are relevant for firm valuation. For all firms with fewer than 10 news stories retrieved by our automatically constructed search queries, we manually search for common names using the Internet and other resources.

Ultimately, our search methods retrieve at least one news story for 1,063 of 1,110 (95.8%) of the firms in the S&P 500 from 1980 to 2004. In addition, we lose another 80 of the 1,063 firms with news stories (7.5%) because these firms did not make the news during the time in which they were in the S&P 500 between 1980 and 2004, which may be quite brief if a firm exits the S&P index shortly after 1980. Also, Factiva's coverage of news stories from 1980 to 1984 appears somewhat incomplete, possibly leading to missing news stories. Finally, after deleting all stories with fewer than three unique positive and negative words or fewer than five total positive and negative words, we lose another three firms, leaving us with 980 qualifying firms. The median firm has 156 news stories, and 929 of 980 firms have at least 10 news stories.

It is possible that we retrieve no news stories for the missing 47 of the initial set of 1,110 S&P 500 firms because of errors in our matching algorithm. Fortunately, although the exact magnitude of our results depends on the matching methodology employed, the sign and significance of all key coefficients does not change for the firms that have been matched using each of the four different processes. Thus, we infer that it is unlikely that matching errors introduce sufficient *systematic* errors in our tests that would significantly change the results. Moreover, our key results depend on cross-sectional and time-series variation in earnings and returns but not the levels of these variables, which could be affected by survivorship bias.

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